

RAMESH KANNAN G

Thoothukudi, Tamil Nadu | +91 9500755440 | rameshkannan8178@gmail.com
[LinkedIn](#) || [GitHub](#)

Python Backend Developer | FastAPI | LLM Integration | RAG Systems

Backend Developer with 1.8 years of experience in building scalable backend Application systems using Python and PHP. Experienced in developing Retrieval Augmented Generation (RAG) pipelines, document processing workflows, and LLM-driven automation tools. Skilled in API design, SQL optimisation, semantic search systems, and production-ready backend architecture.

Technical Skills

Languages: Python, PHP

Frameworks & Libraries: FastAPI, Flask, LangChain, CodeIgniter 4, SpaCy,

Vector Databases: FAISS, Weaviate

Databases: MySQL, PostgreSQL

Tools & Platforms: Git, Postman, Keycloak, Ollama

Professional Experience

Junior Software Developer

MVG Digital Consulting Pvt. Ltd. | Oct 2024 – Present

- Developed and maintained 15+ production REST APIs using FastAPI, CI4 and MySQL supporting internal automation workflows.
- Built LLM-powered semantic search and chatbot modules using LangChain, improving document retrieval efficiency.
- Designed document ingestion pipelines using SpaCy, Regex, and pdf parsing tools for metadata extraction and structured storage.
- Integrated vector search systems (FAISS / Weaviate) to enable contextual keyword matching and similarity-based retrieval.
- Implemented authentication, middleware, and structured error handling for scalable backend services.

- Optimized SQL queries and indexing strategies to improve response latency for high-volume endpoints.
- Collaborated with frontend and product teams to deliver modular backend components aligned with business requirements.

Key Projects

Intelligent Reference Management & Keyword Analysis Tool

- Built an end-to-end Retrieval Augmented Generation pipeline for extracting and classifying research citations.
- Integrated GROBID + NLP preprocessing + embedding storage + LLM inference for automated keyword generation.
- Implemented contextual similarity scoring using vector search to match research articles with citation metadata.
- Designed scalable workflow APIs for ingestion, processing, retrieval, and response generation.

Multi-Discipline Class Management System

- Developed backend modules for scheduling, fee tracking, and notification workflows using PHP (CodeIgniter 4) and MySQL.
- Implemented role-based authentication using Keycloak for secure access control.
- Built notification services including WhatsApp alerts for real-time updates.
- Optimized backend logic and database queries to improve system performance and reliability.

AI System Architecture Exposure

- Designed modular backend pipelines for document upload, preprocessing, embedding generation, vector retrieval & LLM response.
- Implemented metadata filtering and chunking strategies for improving retrieval accuracy.
- Worked on token usage optimisation and prompt structuring for better LLM output quality.

Education

B.E – Mechanical Engineering
Anna University Regional Campus, Madurai | 2020 – 2024
CGPA: 7.8